

Week 3

DevOps Internship

Release Date: 10 August 2026

This week, you'll learn **Shell Scripting, Automation & Web Servers**. These skills help automate repetitive tasks and deploy web applications efficiently in real-world DevOps environments.

Topic:- Shell Scripting, Automation & Web Servers

This week, you are expected to learn:

- Introduction to Bash Shell Scripting
- Variables & User Input
- Conditional Statements
- Loops
- Functions
- File & Directory Operations
- Cron Jobs
- Apache Web Server
- Nginx Web Server
- Basic Web Server Configuration
- Log Management

*Learning Approach

Focus on understanding the logic behind each script and how automation simplifies repetitive tasks. Practice writing your own scripts, deploy a web server, and experiment with different configurations to build confidence.

Keep these points in mind:

- ✓ Write and test every Bash script yourself.
- ✓ Understand the purpose of each command before using it.
- ✓ Practice automating repetitive Linux tasks.
- ✓ Explore Apache or Nginx configuration through hands-on practice.
- ✓ Verify AI-generated scripts before using them in your environment.

*Complete the following practical exercises:

- Write a Bash script to display system information.
- Create a script to automate file backups.
- Develop a script to create users and directories.
- Write a script to clean old log files.
- Schedule a Bash script using Cron.
- Install Apache or Nginx on your Linux system.
- Host a simple HTML webpage.
- Verify that the website is accessible through your browser.
- Prepare a short PDF explaining:
 - Shell Scripting
 - Variables
 - Loops
 - Functions
 - Cron Jobs
 - Apache
 - Nginx
 - Log Management

Task Submission

Once you complete the above task, submit:

- ✓ PDF Report
- ✓ Bash Script Files
- ✓ Screenshot of Script Execution
- ✓ Screenshot of Cron Job
- ✓ Screenshot of Running Website

Task Submission Deadline: Wednesday (EOD)

Scenario

Imagine you have joined a company as a **Junior DevOps Engineer**. Your manager asks you to automate routine Linux tasks and deploy a simple company website on a Linux server.

*Complete the following:

- Create a Bash script to:
 - Create project directories.
 - Generate automatic backups.
 - Copy project files.
 - Compress the backup into an archive.
- Schedule the script using Cron.
- Install Apache or Nginx.
- Deploy a simple HTML website.
- Verify the website is accessible in your browser.
- Document each step performed.

Activity Submission

After completing the hands-on activity, submit:

- ✓ Bash Script Files
- ✓ Screenshot of Cron Job
- ✓ Screenshot of Running Website
- ✓ Screenshot of Backup Archive
- ✓ Short implementation summary

Activity Submission Deadline: Friday (EOD)

AI Assistant Task

Use an AI Assistant such as **ChatGPT, GitHub Copilot, Amazon Q, Gemini, Claude, or Microsoft Copilot** to strengthen your understanding of Shell Scripting and Web Servers.

Complete the following:

- Ask AI to generate a Bash script for automatic backups.
- Ask AI to explain the difference between Apache and Nginx.
- Generate a Shell Scripting cheat sheet with commonly used commands.
- Ask AI to optimize one of your Bash scripts.
- Ask AI to explain how Cron Jobs automate tasks.
- Verify the AI-generated script by testing it on your Linux system.

DevOps Pro Tip

Automate repetitive tasks whenever possible. Even simple Bash scripts can save time, reduce manual effort, and minimize errors in production environments.

Take time to understand the logic behind each script rather than simply copying commands. Strong scripting skills will help you automate repetitive tasks and improve efficiency throughout your DevOps journey.

If you face any genuine issues, feel free to reach out before the submission deadline.